A proof of the conjectured recurrence for OEIS A005558

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Abstract

OEIS A005558 carries a conjectured linear recurrence with polynomial coefficients, of order 3. The entry posts no generating function, and the closed form it does post is not a hypergeometric term in n: it contains a floor, so its shift quotients are not rational and the usual argument does not start. Splitting on the parity of n repairs this. Writing $a(2m) = e_0(m)$ and $a(2m+1) = e_1(m)$, both halves are ordinary hypergeometric expressions in m, and the recurrence becomes two identities in m, since each shifted term $a(n-i)$ lands in whichever half the parity of $n-i$ selects. Both are then decided exactly, by grouping into similarity classes and cancelling. The procedure returns a proof or a refutation; both halves must hold, and here both do.

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1 The sequence and the conjecture

OEIS A005558 is "a(n) is the number of n-step walks on square lattice such that 0 <= y <= x at each step". It has offset 0 and begins

$$a(0), \ldots, a(7) \;=\; 1, 1, 3, 6, 20, 50, 175, 490, \;\ldots$$

The entry states, not as a conjecture,

> a(n) = C(n+1, ceiling(n/2))*C(n, floor(n/2)) - C(n+1, ceiling((n-1)/2))*C(n, floor((n-1)/2)).
> - *Paul D. Hanna*, Apr 16 2004

that is,

$$a(n) \;=\; \binom{n}{\lfloor \frac{n}{2} \rfloor}\binom{n+1}{\lceil \frac{n}{2} \rceil} - \binom{n}{\lfloor \frac{n}{2} - \frac{1}{2} \rfloor}\binom{n+1}{\lceil \frac{n}{2} - \frac{1}{2} \rceil}. \tag{1}$$

This is the only input the proof takes from the entry. It was checked against the entry's own DATA before use, and the index alignment was determined from that check rather than assumed: the formula is stated in the entry's own indexing.

The entry separately carries the following comment:

> **Conjecture:** (n+3)*(n+2)*a(n) - 4*(n^2+3*n+1)*a(n-1) + 16*(-n^2+n+1)*a(n-2) + 64*(n-1)*(n-2)*a(n-3) = 0. - *R. J. Mathar*, Apr 02 2017

As of the "Last modified" line on the live entry (2025-11-10, revision 172) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Written out, the conjecture is

$$(n+2)(n+3)\;a(n) - 4\left(n^2+3n+1\right)\;a(n-1) - 16\left(n^2-n-1\right)\;a(n-2) + 64\,(n-2)\,(n-1)\;a(n-3) \tag{2}$$

2 Hypergeometric terms, and why the closed form is not one

Definition 1. A nonzero expression $c(m)$ is a *hypergeometric term* in m if $c(m+1)/c(m)$ is a rational function of m. Two such terms c, d are *similar*, $c \sim d$, if c/d is a rational function of m.

Products of factorials, binomial coefficients with arguments linear in the variable, exponentials and polynomials are hypergeometric. A floor is not: $\lfloor n/2 \rfloor$ takes the same value at n and $n+1$ for even n and jumps at odd n, so the quotient $c(n+1)/c(n)$ is not a rational function of n, and (1) is therefore outside the reach of the direct argument.

Parity removes the floor. For integer m, $\lfloor 2m/2 \rfloor = m$ and $\lfloor (2m+1)/2 \rfloor = m$, and similarly for the ceiling, so substituting $n = 2m$ and $n = 2m+1$ into (1) and simplifying gives two expressions free of rounding:

$$a(2m) \;=\; e_0(m) \;=\; \frac{\binom{2m}{m}\,(2m+1)!}{(m+1)!^2}, \qquad a(2m+1) \;=\; e_1(m) \;=\; \frac{2\binom{2m+1}{m}\,(2m+1)!}{m!\,(m+2)!}, \tag{3}$$

and each of e_0, e_1 is a sum of finitely many hypergeometric terms in m.

Lemma 2. *Hypergeometric terms that are pairwise dissimilar are linearly independent over the field of rational functions: if $\sum_j u_j(m)c_j(m) = 0$ for rational u_j and all large m, then every u_j is zero. See Petkovšek, Wilf and Zeilberger [1], Chapter 5.*

3 The recurrence, one residue class at a time

Fix the conjectured coefficients $p_0, \ldots, p_r$ and consider $T(n) = \sum_{i=0}^{r} p_i(n)a(n-i)$. Each term reaches a half of (3) chosen by the parity of $n-i$:

Proposition 3. *With $a(2j) = e_0(j)$ and $a(2j+1) = e_1(j)$,*

$$T(2m) = \sum_{i \text{ even}} p_i(2m)\, e_0\big(m - \tfrac{i}{2}\big) + \sum_{i \text{ odd}} p_i(2m)\, e_1\big(m - \tfrac{i+1}{2}\big)\,,$$

$$T(2m+1) = \sum_{i \text{ even}} p_i(2m+1)\, e_1\big(m - \tfrac{i}{2}\big) + \sum_{i \text{ odd}} p_i(2m+1)\, e_0\big(m - \tfrac{i-1}{2}\big)\,.$$

Both right-hand sides are finite sums of hypergeometric terms in m. The recurrence $T(n) = 0$ holds for all large n if and only if both vanish identically, and by Lemma 2 each vanishes identically if and only if, after grouping its summands into similarity classes, every class contributes the zero rational function.

Proof. The two displays are the definition of T with $n-i$ written in the form $2j$ or $2j+1$: for $n = 2m$ the index $n-i$ is even exactly when i is, and for $n = 2m+1$ exactly when i is odd. Every $e_\epsilon(m-s)$ is a sum of hypergeometric terms in m, and $p_i(2m)$, $p_i(2m+1)$ are polynomials in m, so each display is a finite sum of hypergeometric terms. Within a similarity class every summand is the class representative times a rational function, so the class contributes the representative times the sum of those rational functions; the classes are pairwise dissimilar by construction, and Lemma 2 applies. Finally n ranges over both parities, so $T(n) = 0$ for all large n is exactly the conjunction of the two. □

4 The computation for A005558

Carrying out Proposition 3 with the coefficients of (2) and the two halves (3), every similarity class of $T(2m)$ contributes zero, and so does every similarity class of $T(2m+1)$. The even half splits into one class and the odd half into one class.

Theorem 4. *For all $n \geq 3$,*

$$(n+2)(n+3)\ a(n) - 4\left(n^2+3n+1\right)\ a(n-1) - 16\left(n^2-n-1\right)\ a(n-2) + 64\,(n-2)(n-1)\ a(n-3)$$

Proof. Both residue classes vanish identically by the computation above, and every integer n lies in one of them. The excluded values are those at which a class residual has a pole or a term of (1) is undefined. □

5 Verification

Two checks were made, and both are independent of the algebra above.

First, the closed form (1) was evaluated at the first 13 indices and compared term by term with the entry's DATA; the values agree exactly, on both parities. A formula that did not reproduce the entry's own terms would not have been used, and the alignment was fixed by that comparison rather than assumed.

Second, the conjectured recurrence (2) was evaluated on the published terms in exact integer arithmetic at every index where it applies: 25 instances beginning at $n = 3$, all of them zero. This is not part of the proof, which is symbolic, but it would have caught a transcription error in the coefficients.

References

[1] M. Petkovšek, H. S. Wilf and D. Zeilberger, *A=B*, A K Peters, Wellesley MA, 1996.

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A005558, 2025-11-10.

[3] R. L. Graham, D. E. Knuth and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison-Wesley, 1994.